

CERT-IN COMPLIANCE & NATIONAL CYBER RESILIENCE GUIDE

A Statutory Reference and CISO Governance Guide to India's Cybersecurity Architecture

Regulatory Overview: Safeguarding India's rapidly expanding digital landscape—which crossed 100.29 crore wireless connections and processes over 21 billion monthly Unified Payments Interface transactions valued at over ₹27 lakh crore—is a critical national priority. To protect this system, the Union Budget 2025–26 allocated ₹782 crore for cybersecurity, establishing rigorous statutory mandates and sophisticated oversight networks.

1. India's Frontline Cyber Defense Nodal Agency

Operating under the Ministry of Electronics and Information Technology (MeitY) and established under Section 70B of the Information Technology Act, 2000, the Indian Computer Emergency Response Team (CERT-In) is the central nodal agency for national cyber defense. Its broad statutory mandates include the collection, analysis, and dissemination of cyber incident data, forecasting and issuing alerts on emerging threats, coordinating emergency measures to handle active breaches, and issuing formal guidelines, advisories, and security vulnerability notes. In 2025 alone, CERT-In demonstrated its immense operational capability by handling over 29.44 lakh cyber incidents, publishing 1,530 technical alerts, 390 vulnerability notes, 65 advisories, and documenting 29 custom Common Vulnerabilities and Exposures (CVEs) discovered within regional systems.

2. Statutory Incident Reporting Guidelines

Compliance with CERT-In directives is a mandatory legal obligation for all Indian corporate bodies, service providers, and government entities. Under the current regulatory directions, organizations are strictly required to report cybersecurity incidents to CERT-In **within 6 hours** of detection. This mandatory window is designed to enable rapid national-level threat correlation and coordinate defensive responses to limit systemic damage. Furthermore, personal data breaches must be reported under the strict obligations of the Digital Personal Data Protection (DPDP) Act 2023, which mandates immediate notifications to the Data Protection Board and affected users. Additionally, listed commercial entities carry an obligation to disclose material cybersecurity incidents to the BSE or NSE stock exchanges within 24 hours of identification.

3. Specialized Institutional Safeguards & Frameworks

To operationalize cybersecurity policies and coordinate defenses across sectors, CERT-In anchors several specialized institutional frameworks:

Cyber Swachhta Kendra (CSK)

The Cyber Swachhta Kendra (CSK)—the Botnet Cleaning and Malware Analysis Centre—is a massive citizen-centric preventive initiative. Covering 98% of India's digital population, CSK collaborates with internet service providers, industry, and academia to track infected networks and send large-scale alerts to compromised devices. With 1,427 organisations onboarded and 89.55 lakh downloads of its free malware removal tools, CSK plays a vital role in cleaning home routers, mobile systems, and IoT devices.

Sectoral & State CSIRTs

To provide specialized incident response, CERT-In oversees a layered network of Computer Security Incident Response Teams (CSIRTs). Key among these are **CSIRT-Fin** (Financial Sector), which coordinates threat response across banking, insurance, securities, and pension funds, and **CSIRT-Power** (Power Sector), which coordinates proactive containment, security audits, and threat intelligence to secure the national power grid.

National Cyber Coordination Centre (NCCC)

The NCCC provides wide-area situational awareness by scanning internet metadata to detect potential cyber threat vectors. It facilitates real-time information and threat intelligence sharing across central ministries, state governments, and public-private stakeholders.

4. The Anatomy of Cyber Failure Framework

Released in the second edition of the collaborative Digital Threat Report 2025–26 (produced by MeitY, CERT-In, CSIRT-Fin, and SISA), the Anatomy of Cyber Failure framework introduces a <i>4-Layer Gap Archetype Framework</i> to reconstruct how modern security breaches actually occur. This diagnosis recognizes that major breaches are rarely the result of a single security lapse. Instead, they occur due to a chain of compounding weaknesses, such as a combined failure of multi-factor authentication, unsegmented internal network architecture, lack of continuous cloud baseline audits, and outdated incident response runbooks. By mapping controls to this 4-layer model, CISOs can transition from periodic security interventions to continuous risk assessment and direct investments to eliminate systemic vulnerabilities.

5. Security Audits and Capacity Building

Regular audits and simulations are essential to validate enterprise readiness. Under its Security Assurance Framework, CERT-In has empanelled 231 certified IT security audit organisations. These entities conduct independent vulnerability assessments and penetration testing across critical infrastructure networks. To build national capability, CERT-In conducted 122 cybersecurity drills and table-top exercises in 2025 alone, with participation from 1,570 organisations spanning the defense, space, finance, telecommunications, and power sectors. Furthermore, CERT-In conducted 32 specialized technical training programs, educating 20,799 CISO officers and digital safety professionals.

Cybersecurity Compliance and Institutional Mandates

Agency / Program	Statutory Role and Scope	Enterprise/CISO Interface
CERT-In	National nodal agency for incident response, threat alerts, and crisis coordination under IT Act.	Reports incidents within 6 hours; receives technical advisories, vulnerability notes.
CSIRT-Fin	Coordinates cybersecurity incident response and information sharing across the national financial sector.	BFSI entities report incidents; participates in sector threat-sharing platforms.
CSIRT-Power	Coordinates threat intelligence, incident response, and cybersecurity audits across the power sector.	Power utilities report OT anomalies and coordinate critical infrastructure audits.
NCCC	Scans wide-area cyberspace metadata to generate real-time situational awareness and threat alerts.	Receives warning alerts on active threats and localized phishing campaigns.
Cyber Swachhta Kendra	Citizen-centric center providing botnet cleaning and free malware removal tools.	Uses free diagnostic software; registers corporate networks for botnet scanning.
Empanelled Auditors	231 certified security organisations evaluating public and private sector ICT systems.	Enables independent vulnerability assessments, code audits, and pen testing.